

The Dollar Exposure of the G20

Reserve Holdings, Currency Composition, and Monetary Dependence

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Abstract

The United States dollar occupies a central position in the international monetary system and remains the dominant currency in the official foreign exchange reserves of the world. This paper studies the reserve position of the Group of Twenty (G20) economies, with particular emphasis on the amount of foreign exchange reserves held by each economy, the estimated share of those reserves denominated in U.S. dollars, and the resulting degree of dollar dependence.

A central empirical difficulty is that the International Monetary Fund's Currency Composition of Official Foreign Exchange Reserves (COFER) database does not disclose country-level currency allocations. Country-level COFER observations are confidential. We therefore distinguish carefully between observed total reserve assets and estimated or disclosed currency composition. Rather than treating unavailable observations as zero, the paper develops a transparent classification of country-level USD exposure into directly disclosed, independently estimated, and non-identifiable components.

The paper reports descriptive statistics for the G20, including the mean, median, standard deviation, minimum, maximum, coefficient of variation, reserve-weighted USD share, and measures of cross-country heterogeneity. It additionally considers reserves relative to gross domestic product, imports, and short-term external liabilities. The analysis distinguishes between USD reserve exposure and broader dollar dependence, arguing that the latter also incorporates trade invoicing, external liabilities, financial intermediation, and international funding.

The paper concludes by considering the implications of reserve concentration for central-bank portfolio management, monetary sovereignty, financial stability, and institutional design.

The paper ends with “The End”

Keywords: U.S. dollar, foreign exchange reserves, G20, reserve currencies, central banks, monetary dependence, international monetary system, reserve management, systemic risk.

1 Introduction

The U.S. dollar occupies a uniquely important position in the international monetary and financial system. Central banks hold dollar-denominated assets as part of their official foreign exchange reserves; international trade is frequently invoiced in dollars; global financial contracts are extensively denominated in dollars; and international borrowers and financial institutions frequently obtain dollar funding.

This paper asks a relatively simple empirical question:

How dependent are the G20 economies upon the U.S. dollar through their official foreign exchange reserves?

The question is deceptively difficult.

A country’s total international reserves can generally be observed. However, the currency composition of those reserves is substantially less transparent. The IMF’s Currency Composition of Official Foreign Exchange Reserves (COFER) database provides the global composition of official foreign exchange reserves, but country-level currency allocations are confidential.

Consequently, an empirical analysis of the G20 must distinguish between two objects:

$$R_i = \text{total official reserve assets of country } i,$$

and

$$D_i = \text{U.S.-dollar-denominated foreign exchange reserves of country } i.$$

The corresponding dollar share is

$$s_i^{USD} = \frac{D_i}{FX_i},$$

where FX_i denotes foreign exchange reserves rather than total reserve assets including monetary gold, Special Drawing Rights (SDRs), and reserve positions in the IMF.

The distinction matters.

A country can possess a large stock of reserves while maintaining a relatively diversified currency portfolio. Conversely, a country can have moderate reserves but a very high dollar concentration.

The purpose of this paper is therefore not merely to rank countries by reserve holdings. It is to construct a statistical description of the distribution of dollar exposure across the G20.

2 The Data Problem

2.1 COFER

The IMF COFER dataset reports the currency composition of official foreign exchange reserves at the aggregate level. The currencies separately identified include the U.S. dollar, euro, Chinese renminbi, Japanese yen, pound sterling, Australian dollar, Canadian dollar, and Swiss franc.

The IMF conducts the COFER survey quarterly. Country-specific observations are confidential, and the publicly disseminated dataset does not provide country-level currency composition.

Beginning in 2025Q3, the IMF revised COFER and eliminated the previous “unallocated” category through statistical imputation. The revised historical series extends this methodology back to 2000Q1.

This development is important for the present paper because it creates a more complete global benchmark while preserving country-level confidentiality.

2.2 Country-Level Reserve Data

For country-level reserve stocks we use the IMF International Reserves and Foreign Currency Liquidity (IRFCL) framework where available.

Let

$$RA_i$$

denote total reserve assets,

$$G_i$$

denote monetary gold,

$$SDR_i$$

denote SDR holdings,

$$RP_i$$

denote the reserve position in the IMF, and

$$FX_i$$

denote foreign exchange reserves.

Conceptually,

$$RA_i = G_i + SDR_i + RP_i + FX_i.$$

The present analysis concentrates on FX_i when constructing a currency composition measure.

3 G20 Sample

The empirical sample consists of the G20 economies that are individual countries:

Country	Abbreviation
Argentina	ARG
Australia	AUS
Brazil	BRA
Canada	CAN
China	CHN
France	FRA
Germany	DEU
India	IND
Indonesia	IDN
Italy	ITA
Japan	JPN
Mexico	MEX
Republic of Korea	KOR
Russia	RUS
Saudi Arabia	SAU
South Africa	ZAF
Türkiye	TUR
United Kingdom	GBR
United States	USA

The European Union and African Union are institutional members of the G20, but are not treated as additional countries in the country-level sample. The United States is retained because it is itself a G20 member, although its reserve position requires special interpretation given the dollar's role as the principal reserve currency.

4 Measurement of Dollar Exposure

4.1 Directly Observed USD Holdings

Where a central bank or monetary authority publicly reports the currency composition of its reserves, we define

$$D_i^{obs}$$

as the directly observed USD component.

The corresponding observed share is

$$s_{i,obs}^{USD} = \frac{D_i^{obs}}{FX_i}.$$

4.2 Estimated USD Holdings

Where direct country-level information is unavailable, an estimated quantity may be constructed from publicly available information.

We denote this quantity by

$$\hat{D}_i.$$

The estimated dollar share is

$$\hat{s}_i^{USD} = \frac{\hat{D}_i}{FX_i}.$$

The paper does not treat $\hat{D}_i$ as an observed fact. It is an estimate and is therefore accompanied by an uncertainty classification.

4.3 Three-Tier Evidence Classification

We classify each country observation into three categories:

1. directly disclosed currency composition;
2. independently estimated currency composition;
3. currency composition not sufficiently identifiable.

This prevents an important statistical error:

missing USD composition $\neq$ zero USD holdings.

Where the USD share cannot be credibly established, the observation is treated as missing rather than assigned an artificial value.

5 Descriptive Statistics

Let

$$s_1, s_2, \dots, s_N$$

denote the country-level USD shares for the set of observations for which a sufficiently credible estimate exists.

The arithmetic mean is

$$\bar{s} = \frac{1}{N} \sum_{i=1}^N s_i.$$

The median is

$$\tilde{s} = \text{median}(s_1, \dots, s_N).$$

The sample variance is

$$\hat{\sigma}^2 = \frac{1}{N-1} \sum_{i=1}^N (s_i - \bar{s})^2.$$

The standard deviation is

$$\hat{\sigma} = \sqrt{\hat{\sigma}^2}.$$

The coefficient of variation is

$$CV = \frac{\hat{\sigma}}{\bar{s}}.$$

We additionally report the minimum and maximum:

$$s_{\min} = \min_i s_i,$$

and

$$s_{\max} = \max_i s_i.$$

6 Unweighted versus Reserve-Weighted Dollar Exposure

A central distinction of the paper is between the average country and the average dollar held in G20 reserves.

The unweighted average is

$$\bar{s} = \frac{1}{N} \sum_i s_i.$$

However, the reserve-weighted USD share is

$$S_{USD}^W = \frac{\sum_i D_i}{\sum_i FX_i}.$$

These quantities need not be equal.

If a small number of very large reserve holders have exceptionally high or low dollar shares, then

$$S_{USD}^W \neq \bar{s}.$$

This distinction is economically important because the unweighted statistic answers the question

What does the typical G20 country look like?

whereas the weighted statistic answers

What fraction of the G20 reserve stock is exposed to the dollar?

7 Reserve Adequacy and Dollar Exposure

Dollar exposure should also be considered relative to the economic scale of each country.

Define reserves relative to GDP as

$$RGDP_i = \frac{FX_i}{GDP_i}.$$

Dollar reserves relative to GDP are

$$DRGDP_i = \frac{D_i}{GDP_i}.$$

An economy with a relatively small dollar share but enormous reserves relative to GDP may have greater absolute dollar exposure than an economy with a larger dollar share but much smaller reserves.

We therefore define a simple dollar-reserve intensity measure:

$$DRI_i = s_i^{USD} \frac{FX_i}{GDP_i}.$$

Equivalently,

$$DRI_i = \frac{D_i}{GDP_i}.$$

This measure captures the dollar reserve position relative to the domestic economic base.

8 Reserve Concentration

The concentration of the G20 reserve stock can be studied using the Herfindahl–Hirschman Index.

Let

$$w_i = \frac{FX_i}{\sum_j FX_j}.$$

Then

$$HHI = \sum_i w_i^2.$$

A high value indicates that the aggregate G20 reserve stock is concentrated among a small number of reserve holders.

A second concentration measure can be constructed for dollar reserves:

$$w_i^{USD} = \frac{D_i}{\sum_j D_j},$$

giving

$$HHI_{USD} = \sum_i (w_i^{USD})^2.$$

Comparing

$$HHI_{USD}$$

with

allows us to distinguish general reserve concentration from specifically dollar-related concentration.

9 The United States as a Special Observation

The United States requires special treatment.

The dollar is the dominant reserve currency internationally, but the United States does not need to hold foreign currency reserves in the same sense as an economy whose domestic currency is not the principal international reserve currency.

Consequently, including the United States in an unadjusted country-level USD-share calculation can produce a misleading interpretation.

We therefore report two samples:

1. the full 19-country G20 sample;
2. the G20 sample excluding the United States.

The second specification is particularly useful when the objective is to measure the dollar dependence of economies that use a currency other than the dollar as their domestic currency.

10 Global Benchmark

The global COFER aggregate provides an external benchmark against which the G20 results can be compared.

For 2026Q1, global official foreign exchange reserves were approximately

13.10 trillion USD,

while the U.S. dollar represented approximately

57.13%

of global foreign exchange reserves.

Thus,

$$D_{world} \approx 0.5713(13.10) \approx 7.48 \text{ trillion USD.}$$

This figure is an aggregate global benchmark and should not be interpreted as the sum of publicly observable country-level USD holdings.

The distinction is essential because the IMF's global COFER estimate uses statistical aggregation and imputation while maintaining the confidentiality of individual country observations.

11 Expected Empirical Results

The empirical analysis will produce a table of the following form.

Table 1: G20 Foreign Exchange Reserves and USD Exposure

Country	FX Reserves (\$ bn)	USD Reserves (\$ bn)	USD Share (%)	FX/GDP (%)	USD/GDP (%)
Argentina	—	—	—	—	—
Australia	—	—	—	—	—
Brazil	—	—	—	—	—
Canada	—	—	—	—	—
China	—	—	—	—	—
France	—	—	—	—	—
Germany	—	—	—	—	—
India	—	—	—	—	—
Indonesia	—	—	—	—	—
Italy	—	—	—	—	—
Japan	—	—	—	—	—
Mexico	—	—	—	—	—
Korea	—	—	—	—	—
Russia	—	—	—	—	—
Saudi Arabia	—	—	—	—	—
South Africa	—	—	—	—	—
Türkiye	—	—	—	—	—
United Kingdom	—	—	—	—	—
United States	—	—	—	—	—

The completed dataset should report not only point estimates but also the source and evidence classification for every USD-share observation.

12 Statistical Summary

The principal descriptive-statistics table is

Table 2: Descriptive Statistics of G20 USD Reserve Shares

Statistic	Value
Number of observations	—
Mean	—
Median	—
Standard deviation	—
Minimum	—
Maximum	—
First quartile	—
Third quartile	—
Interquartile range	—
Coefficient of variation	—
Reserve-weighted USD share	—

The difference between the arithmetic mean and the reserve-weighted share will be treated as an economically meaningful statistic rather than as a mere mathematical curiosity.

13 USD Reserve Exposure and Dollar Dependence

Official reserve holdings are only one component of dollar dependence.

Define total dollar dependence as

$$DD_i = f(D_i, Trade_i^{USD}, Debt_i^{USD}, Funding_i^{USD}, Banking_i^{USD}, Derivative_i^{USD}).$$

The paper deliberately distinguishes

USD Reserve Exposure $\neq$ Total Dollar Dependence.

A country may diversify its reserve portfolio while remaining highly dependent upon the dollar through international trade, external debt, banking, derivative markets, and international funding.

This distinction is particularly important for emerging-market economies.

14 Reserve Currency Diversification

Suppose a country allocates its foreign exchange reserves across K currencies.

Let

$$p_{ik}$$

be the share of country i 's foreign exchange reserves denominated in currency k .

A currency-concentration index can then be defined as

$$CI_i = \sum_{k=1}^K p_{ik}^2.$$

The corresponding effective number of reserve currencies is

$$ENC_i = \frac{1}{CI_i}.$$

A country whose reserves are concentrated entirely in one currency has

$$ENC_i = 1.$$

A perfectly diversified portfolio across K currencies has

$$ENC_i = K.$$

This provides a natural way to distinguish reserve size from reserve diversification.

15 The Dollar and Systemic Monetary Risk

A dominant reserve currency creates both benefits and risks.

The benefits include liquidity, deep capital markets, low transaction costs, and access to a large international financial ecosystem.

The risks arise from concentration.

If a country's reserve portfolio, trade invoicing, external debt, and financial system are simultaneously dollar-dependent, then a shock to the dollar or to dollar funding conditions can propagate through several channels.

We may represent this schematically as

Dollar Shock $\rightarrow$ Reserve Valuation $\rightarrow$ Exchange Rate $\rightarrow$ External Debt $\rightarrow$ Financial Conditions $\rightarrow$ Domestic Economy.

The strength of this transmission is likely to differ substantially across countries. This heterogeneity is one of the principal empirical objects of the paper.

16 Robustness and Uncertainty

Because country-level currency composition is not universally observable, the analysis should not rely upon a single estimated number.

For countries for which USD shares are estimated, define

$$s_i^{USD} \in [L_i, U_i],$$

where L_i and U_i represent a plausible uncertainty interval.

The corresponding aggregate USD exposure then satisfies

$$\sum_i L_i F X_i \leq D_{G20} \leq \sum_i U_i F X_i.$$

This interval approach is preferable to false precision.

The paper will therefore report:

1. point estimates where defensible;
2. ranges where uncertainty is material;
3. missing observations where estimation would be unjustified.

17 Institutional Implications

The empirical results have direct implications for reserve management.

A central bank does not need to choose between

“hold dollars”

and

“do not hold dollars.”

The economically relevant problem is portfolio optimization under liquidity, solvency, exchange-rate, geopolitical, and systemic constraints.

Let

$$x_{ik}$$

denote the fraction of reserves held in currency k .

The reserve manager solves a problem of the form

$$\max_{\mathbf{x}_i} U_i(\mathbf{x}_i)$$

subject to

$$\sum_k x_{ik} = 1,$$

$$x_{ik} \geq 0,$$

and appropriate liquidity and risk constraints.

A generalized risk functional may be written

$$R_i = \alpha V_i + \beta L_i + \gamma C_i + \delta G_i,$$

where V_i represents valuation risk, L_i liquidity risk, C_i concentration risk, and G_i geopolitical or systemic risk.

The optimal reserve composition therefore depends upon the central bank's institutional objective rather than upon a universally optimal currency share.

18 Austria as a Case Study

Austria provides an especially interesting institutional case because its monetary environment is embedded within the euro area.

Consequently, an Austrian reserve institution does not face exactly the same reserve-management problem as a central bank whose domestic currency is not itself a major reserve currency.

The Austrian problem is therefore not simply

How many dollars should Austria hold?

but rather

How should Austria manage its exposure to the international monetary system given that its domestic monetary environment is the euro area?

This is an institutional-design question.

The answer could involve diversification, liquidity management, systemic risk management, or alternative reserve architectures.

The empirical G20 analysis provides a benchmark against which such institutional choices can be evaluated.

19 Conclusion

The U.S. dollar remains the dominant currency in the international reserve system. The relevant empirical question, however, is not simply whether the dollar is dominant.

The more useful question is:

How is dollar exposure distributed across countries, and what does that distribution imply for monetary and financial-system resilience?

The G20 provides a natural laboratory because its members include the world's largest advanced and emerging economies and encompass highly diverse exchange-rate regimes, financial systems, reserve positions, and geopolitical circumstances.

The empirical analysis developed in this paper therefore has three objectives.

First, it measures the size of G20 foreign exchange reserves.

Second, it estimates the USD component wherever sufficiently reliable evidence permits.

Third, it measures the statistical distribution and concentration of dollar exposure.

The most important methodological conclusion is that unavailable country-level COFER observations must not be confused with zero USD holdings. The paper consequently treats the currency-composition problem as one of partial observability.

The resulting framework allows us to distinguish

reserve size,

USD concentration,

and

broader dollar dependence.

Those are three different economic concepts.

The international monetary system can therefore be studied not only as a system of currencies, but as a system of interconnected balance sheets in which reserve managers, governments, financial institutions, firms, and households are exposed to common monetary infrastructure.

That perspective provides a natural bridge between reserve management, international finance, systemic risk, and institutional design.

Data Sources

The principal data sources for the empirical implementation are:

1. International Monetary Fund, Currency Composition of Official Foreign Exchange Reserves (COFER).
2. International Monetary Fund, International Reserves and Foreign Currency Liquidity (IR-FCL).
3. International Monetary Fund, World Economic Outlook and associated macroeconomic databases.
4. World Bank, World Development Indicators.
5. National central banks and monetary authorities, where country-level currency composition is publicly disclosed.

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The End